Baslan Asaad

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EDUCATION

Rutgers University Honors College **Piscataway, NJ**
B.S. in Electrical Engineering & B.S. in Computer Science | Expected Graduation: May 2028 | GPA: 3.96/4.00

SKILLS

Languages & Software: Python, C++, Java, JavaScript, SQL, REST APIs, Full-Stack Web, Multithreading, Git / GitLab, Linux
AI & Vision: Machine Learning / Computer Vision, PyTorch, OpenCV, LLMs, Transformers, Embeddings, Synthetic Data
Robotics: ROS, Autonomous Systems, PID / Control Loops, SLAM (LIO-SAM), Lidar, Sensor Fusion
Hardware: PCB Design (KiCad), Circuit Simulation (LTspice), SolidWorks, Arduino, Raspberry Pi, Serial Communications

EXPERIENCE

EV-Edison **Kearny, NJ**
Software & Electrical Engineering Intern *May 2026 – Aug. 2026*

- Architected and developed EV-Edison's official website from scratch, growing traffic up to 1,100+ weekly unique visitors (see ev-edison.com or baslan.dev/ev-edison).
- Incorporated targeted SEO architecture, dynamic location-clustering maps, video-free performant hero animations, and automated contact pipelines via Resend API.
- Designed contractor-approved electrical infrastructure layouts for 6 EV sites under construction, optimizing transformer and charger placement to minimize trenching and validating spatial clearance via vehicle simulation.
- Developed an internal software tool to automate the EV site design process, accelerating layout generation and electrical placement workflows.

Columbia University CS3 Accelerator **New York, NY**
Technical Lead *Oct. 2026 – Present*

- Developing an autonomous routing system to find the lowest-risk paths for commercial fleets and public users, unlocking dynamic insurance data that reduces premiums and redefines road liability.
- Leading customer discovery across 20+ key industry stakeholders to refine technical product requirements and develop a scalable business model.

General Autonomy **Piscataway, NJ**
Software & Mechatronics Engineering Intern *Sep. 2025 – Present*

- Developing control software for autonomous test vehicles to execute high-speed transient drift maneuvers under dynamic conditions.
- Designing and building electromechanical systems for a robotic drive-by-wire module that remotely actuates steering, throttle, and braking.
- Designed and implemented a lidar-embedded Taxi Sign payload to enable LIO-SAM 3D mapping in dense urban environments.

Falcon Attack and Defense Autonomy **Little Falls, NJ**
Aerospace & Software Engineering Intern *May 2025 – Aug. 2025*

- Developed GPS-denied proportional navigation from scratch for high-speed drones, using camera-based computer vision to achieve reliable kinetic interception of moving targets.
- Built comprehensive debug software tools, including radar visualization, video-stream banks, and 3D real-time flight path reconstruction.
- Developed 2 synthetic data generators and full-stack web application features to streamline the simulation, training, and validation of target detection Markov models.
- Managed GitLab version control workflows, reviewing MRs across 5 contributors and guiding interns through integration.

PROJECTS (see https://baslan.dev for more)

Generative AI Product Classifier

- Built an AI-powered product classification system using embeddings and LLMs for hierarchical taxonomy prediction, and optimized semantic search using transformer-based models to improve scalability.

Ultraviolet Sanitation Project

- Designed and built a UV-C sanitation system for reducing viral contamination in shared environments, and deployed it in community settings while presenting the project to local organizations.